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An efficient intelligent data fusion algorithm for wireless sensor network

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Abstract

Wireless sensor network (WSN) are usually restricted by assembled batteries which are difficult to recharge, therefore saving network energy is crucial for WSN. For increasing the survival time of the network, an efficient intelligent data fusion algorithm named GAPSOBP is put forward which integrating BP neural network, genetic algorithm and particle swarm optimization algorithm reasonably. In GAPSOBP, wireless sensors are analogy to neurons in the neural network. Data collected by sensors is extracted by BP neural network, and then combined with clustering routing to fuse extra data, thus reducing data volume sent to base station or sink node. Simulation results show that GAPSOBP is superior than LEACH and PSOBP algorithms in terms of energy consumption and network lifetime.

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Keywords: Wireless sensor networks, data fusion, neural network, genetic algorithm, Particle swarm optimization;

1. Introduction

Wireless sensor network is a distributed information network composed of a lot of sensor nodes. Each sensor node can perceive and collect the external environment data, which is retransmitted to the sink node for further processing in a certain routing way. Thanks to the microelectronic and information technologies, the production cost of sensors is reduced, and the types of sensors are increased. In WSN, each node completes the communication and data transmission in a wireless way. Because sensor nodes are usually powered by their own batteries, the power is limited, once the energy is exhausted, the nodes can not continue to work, the robustness of the network structure

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will be affected, leading to the decline of the overall performance of WSN. The energy consumption in WSN mainly involves two stages: data acquisition and data transmission. Min R pointed out that the energy consumption in the data transmission phase is far greater than that in the data acquisition phase, and the energy consumed in transmitting 1 byte data can be used to perform thousands of sensing tasks [2].

Data fusion technology can extract the data characteristics of the original data and conduct comprehensive analysis and processing. It can not only reduce the amount of redundant data, but also improve the reliability of data. It is very suitable for wireless sensor network in battlefield environment. Thus, the introduction of data fusion technology into wireless sensor network for optimal processing of sensing data can effectively reduce data redundancy and network traffic, thus saving node energy and prolonging network lifetime [3].

For reducing energy consumption caused by data transmission, various data fusion algorithms are put forward to cut down data volume transmitted in WSN. For example, Li Haiyan proposed the data fusion method based on Kalman filter [4], which fuses the original sensing data, but the fusion accuracy is low; Lian Fangyuan proposed an improved WSN data fusion algorithm based on BP neural network [5] which can reduce the data redundancy and shorten the data fusion process time. However, the selection of primary weights is difficult. PSO-BP based data fusion method proposed by Chen Qihong can effectively avoid the dilemma of network parameters selection and speed up the data fusion speed[6]. However, it can only obtain the sub-optimal solution. In this paper, aiming at the shortcomings of existent research works on data fusion methods in WSN, a comprehensive intelligent data fusion algorithm named GAPSOBP is proposed. GAPSOBP algorithm introduces often used genetic algorithm operations into the optimization process of PSO algorithm, enriches the optimization space and prevents the solution falling into partial optimization to the greatest extent. The optimized PSO algorithm is used to obtain the initial parameters of BP neural network, speeding up the convergence process and improving data fusion accuracy.

2. GAPSOBP Algorithm Design

2.1. System Model

Firstly, classic LEACH algorithm is adopted as the network routing algorithm [10]. The basic idea is to cluster the wireless network in a round, and generate a cluster head in each cluster to receive data collected by the member nodes of the cluster and transmit fused data to the sink node. Each data transmission procedure can be divided into two stages: the establishment of clusters and the stabilization of data transmission. Since cluster head nodes consume more energy than that of member nodes in cluster because more data transmission processing is needed in cluster heads. In LEACH protocol, for balancing the energy consumption, the cluster head will be selected again after the end of a round of running [11]. The nodes selected as cluster heads in one round will no longer become cluster heads in the next round; on the contrary, with the increase of the running rounds of the protocol, the probability of becoming cluster heads increases. The network model depicted above is given in Figure 1.

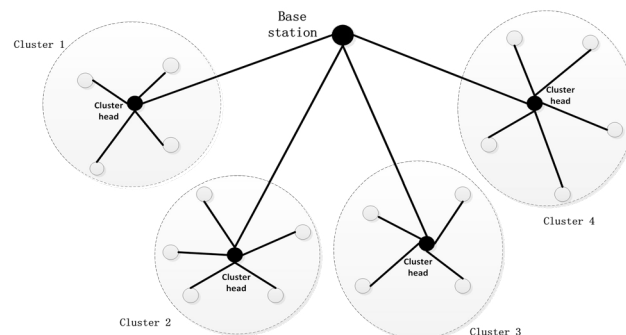


Fig. 1. WSN model based on LEACH.

2.2. Design Methods

In GAPSOBP algorithm, the first task is to determine neural network structure in terms of the WSN topology [12]. Cluster structure is correspondent to BP neural network structure and cluster member nodes is seen as input layers. In addition, this neural network has only one output layer. The number of hidden layers can be calculated by formula 1.

$$l = \sqrt{(m+n)} + a \quad (0 < a < 10) \quad (1)$$

In formula 1, m and n are the numbers of output layers and input layers respectively. The working procedure of GAPSOBP can be depicted as the following three main stages shown in Figure 2.

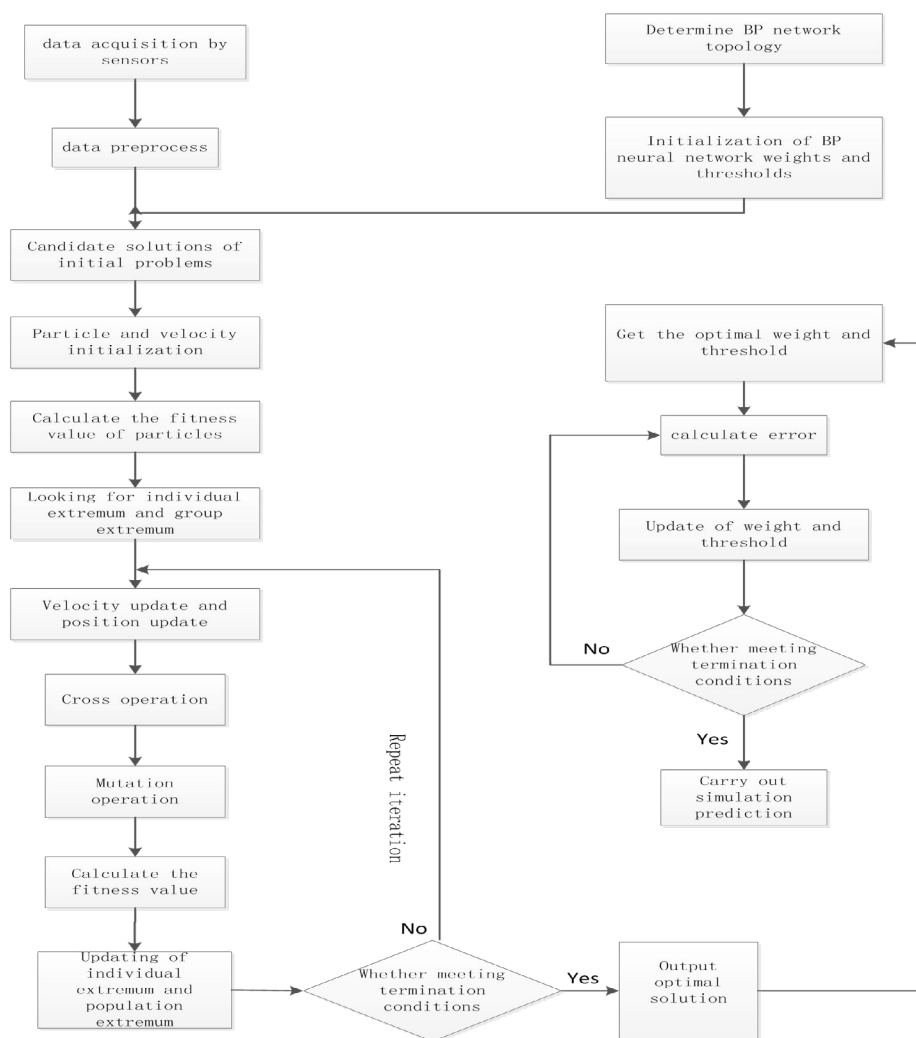


Fig. 2. Working flowchart of GAPSOBP

1) Initializing

In initiation, the neural network structure can be set firstly. Then the initial weights and thresholds are selected

and transferred to PSO algorithm to obtain the primary population. PSO encodes the weights and thresholds in real number, initializes the candidate solution and particle velocity.

2) Setting parameters

The PSO algorithm calculates extreme values for individual and population and updates PSO particle position and speed to speed up the optimization process. In addition, introducing crossover and mutation operations into the PSO optimization process. When the termination conditions are satisfied, the optimization results are used for training by BP neural network. Or, when the preset running time is reached the algorithm is terminated.

3) Training network

The optimization results of genetic PSO are adopted by BP neural network for training network, so as to obtain the network parameters for performing the data fusion processing of clustering network structure.

When each cluster uses the GAPSOBP algorithm for data fusion, the nodes in the cluster collect the data. GAPSOBP uses the sensing data to train the BP neural network. After performing necessary preprocessing, the data collected by member nodes is sent to the cluster head. GAPSOBP algorithm is used for data fusion, and the cluster head transmits the fusion results to the sink node.

3. Implementation of GAPSOBP

Based on above working flow, the implementation of GAPSOBP are explained as follows:

Step 1: According to the number of neurons, required parameter must be initialized, including weights ω_{ij}, ω_{jk} , and threshold values α_j and β_k ;

Step 2: The connection weights ω_{ij} and ω_{jk} , and threshold values α_j and β_k , are transferred to PSO algorithm. Then the particle optimization initial speed is carried out based on preset coding;

Step 3: The fitness value of particle is calculated, searching the extreme values of individual population, and then the velocity and position of particles are recalculated by using the following formulas:

$$V_{is}(t+1) = \omega V_{is}(t) + c_1 r_1 (P_{is}(t) - X_{is}(t)) + c_2 r_2 (P_{gs}(t) - X_{is}(t)) \quad (2)$$

$$X_{is}(t+1) = X_{is}(t) + V_{is}(t+1) \quad (3)$$

In the above formula, V_{is} is the velocity of particle i in the s -dimension direction, ω is the inertia weight to avoid the algorithm falling into the local optimum setting. r_1 and r_2 is real number between (0,1), c_1 and c_2 is two nonnegative constants to ensure the local optimization ability of PSO algorithm. X_{is} represents the position of particle i and t is the current iteration number.

Step 4: Select particles for performing crossover and mutation operations, calculating fitness value, searching individual and population extremum, judge whether the conditions are satisfied, go to step 5; otherwise, return to step 3;

Step 5: Transfer the results of PSO to BP neural network, and obtain e_k according to predicted output Y and O as shown in formula (4):

$$e_k = O_k - Y_k \quad (k = 1, 2, \dots, m) \quad (4)$$

Step 6: Update the threshold and weights in BP neural network as follows;

$$\omega_{ij} = \omega_{ij} + \eta H_i (1 - H_i) X(i) \sum_{k=1}^m \omega_{jk} e_k \quad (i = 1, 2, \dots, n; j = 1, 2, \dots, l; k = 1, 2, \dots, m) \quad (5)$$

$$\omega_{jk} = \omega_{jk} + \eta H_j e_k \quad (j = 1, 2, \dots, l; k = 1, 2, \dots, m) \quad (6)$$

$$\alpha_j = \alpha_j + \eta H_j (1 - H_j) \sum_{k=1}^m \omega_{jk} e_k \quad (j=1,2,\dots,l) \quad (7)$$

$$\beta_k = \beta_k + e_k \quad (k=1,2,\dots,m) \quad (8)$$

Step 7: If output results exceed thresholds or reaching iteration time, the algorithm is completed; otherwise, go to step 5.

When GAPSOPB is used in data fusion for clustering wireless sensor networks, firstly, a clustering network is constructed according to LEACH protocol, and then the relevant weights and thresholds are selected according to the specific network structure. In wireless sensor network, considering saving energy consumption, the complex optimization training procedure can be carried out in the sink node with abundant energy. After the BP neural network training is completed, the sink node transmits the relevant parameters to the corresponding clusters through special messages, and each cluster performs data fusion processing in term of above system network model, then cluster head can transmit the fused data to the sink node.

4. Simulation Experiments

In the simulations, node energy consumption, the amount of data received by the sink node, the survival time of nodes and the iteration time of algorithms are selected as the performance indicators. The performance of LEACH, PSOBP and GAPSOPB algorithms in WSN is compared and analyzed in the following.

For proving the effectiveness of GAPSOPB algorithm, the energy consumption of WSN networks running different algorithms is simulated and compared. Figure 3 shows the specific statistical results.

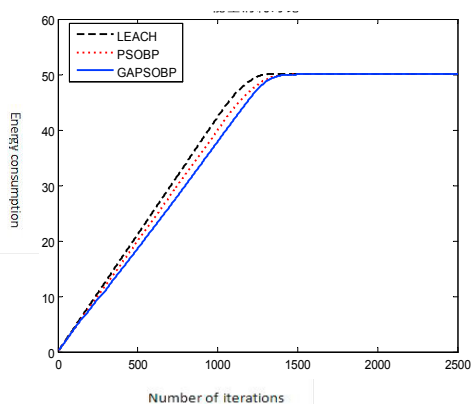


Fig. 3. Comparison of energy consumption

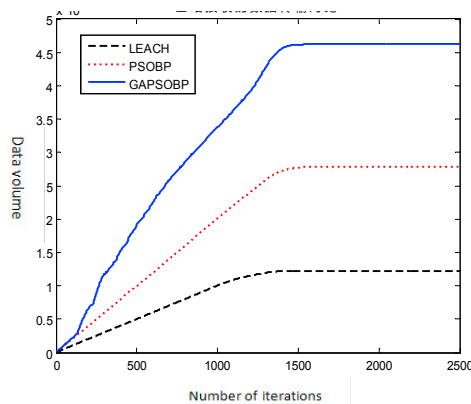


Fig. 4. Comparison of received data volume

As shown in Figure 3, the energy consumption of the network running LEACH is faster than the network running PSOBP algorithm, while the energy consumption of the latter is faster than the network running GAPSOPB algorithm. The reason is PSOBP algorithm can effectively fuse sensor network data, reduce data transmission and save node energy compared with LEACH protocol; however, due to the lack of GA algorithm in PSO for optimization, there is a shortage of long training time and error in fusion results, and the data fusion effect is not as good as GAPSOPB. It is proved that GAPSOPB algorithm is more effective than the former two methods in data fusion, reducing the network data transmission and energy consumption.

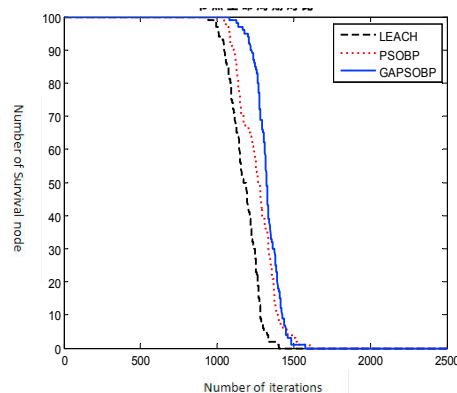


Fig. 5. Comparison of network lifetime

Next, the number of packets received by sink node are compared and analyzed. The specific statistical results can be seen in Figure 4. In Figure 4, the sink node receives less data in the network running LEACH algorithm. Because in the network running LEACH algorithm, with the increase of the amount of data collected by nodes, each node can send more data to cluster head nodes. Cluster heads collect a large number of data packets and need to send them to sink nodes, which will cause network congestion soon. A large amount of data will be discarded and retransmitted. Finally, the useful packets received by sink nodes will be limited. PSOBP effectively fuses Part of data, reduces network traffic, eliminates part of congestion, and improves channel utilization; however, due to the lack of variability in the optimization process of PSO algorithm, data fusion of BP neural network is not sufficient. GAPSOBP algorithm uses genetic particle swarm optimization in the parameter setting of BP neural network, leading to better overall solutions, thus increasing the data fusion degree of BP neural network, greatly reducing the transmitted data volume, improving the effectiveness of wireless sensor network data process.

In this paper, the network lifetime is relevant to node energy depletion directly[13]. In figure 5 the network lifetime comparison results of three different algorithms are given. As we see, when node number is increased to 100, the wireless sensor network running LEACH perform insufficient data fusion and the node energy consumption is too large, which leads to the node death when the algorithm iterations to about 1000 times. The wireless sensor network running GAPSOBP algorithm can effectively fuse part of the sensing data and reduce node energy depletion to a certain extent. WSN with PSOBP data fusion algorithm will not appear until about 1200 seconds. The wireless sensor network using GAPSOBP data fusion algorithm has better data fusion degree, lower energy consumption speed and longer node life cycle. Dead nodes begin to appear after 1300 iterations, which proves that GAPSOBP algorithm can prolong the network lifetime more than the other two algorithms.

5. Conclusions

GAPSOBP combines the traits of traditional GA and BP algorithms into WSN, and reasonably applies PSO algorithm for data fusion processing. GAPSOBP optimizes the initial parameters of BP network by using genetic PSO algorithm, reduces the convergence time of BP neural network, and enhances the data transmission performance apparently. Compared with the traditional LEACH and PSOBP algorithms, GAPSOBP algorithm is better than the former two algorithms in the degree of data fusion, which decreasing the transmitted data volume, saving energy consumption, and better achieving the purpose of extending the network lifetime.

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